

YOONA LI

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EDUCATION

University of Pennsylvania, Penn Engineering
Master of Applied Science in Computer Science (MAS-CS)
New York University
Bachelor of Economics & Statistics

Philadelphia, PA
Expected May 2028
New York, NY
May 2026

TECHNICAL & RESEARCH SKILLS

Programming & Data: Python, SQL, R, MATLAB; NumPy/Pandas, data wrangling, data pipelines, statistical analysis, reproducible experimentation
ML / Deep Learning: PyTorch, scikit-learn, XGBoost, Random Forest; classification, fine-tuning, feature engineering, model evaluation, model explainability (SHAP, LIME), latency/accuracy benchmarking
LLM / GenAI: RAG, prompt engineering, LLM evaluation, LoRA/LoRA+/QLoRA, retrieval-aware generation and abstention, groundedness/uncertainty evaluation, LiveKit/TTS, stateful conversational systems
Research / Evaluation: experimental and survey design, annotation/QC framework design, behavioral-data evaluation, mixed methods, small-N analysis, reproducibility, error analysis

SELECTED AI / ML RESEARCH & PROJECTS

Lumi AI *Jun 2026 - Present*
Founder & Technical Product Lead

- Architected a stateful RAG-based conversational AI system connecting teacher-defined objectives and uploaded knowledge to child-facing interaction, with retrieval-aware abstention, configurable knowledge boundaries, parent-facing evidence, and real-time voice via LiveKit/TTS.
- Designed Test & Compare evaluation workflows spanning retrieval, prompt, and response configurations, measuring groundedness, uncertainty calibration, age-appropriate behavior, evidence of understanding, and response quality.
- Structured retrieval, generation, safety, and evaluation as modular components to support controlled experimentation and reproducible comparison across model, prompt, and grounding configurations.

FashionVote *Aug 2025 - Dec 2025*
Founder & Research Lead

- Built and deployed an AI-driven fashion analytics platform using DeepFashion imagery and a crowdsourced pairwise-voting interface; curated and deduplicated image pairs and trained Random Forest models achieving 90%+ accuracy in trend classification.
- Analyzed preference data from 10K+ users and diagnosed sampling bias and label sparsity to refine model interpretation; attracted retail partnerships exploring marketing and product-design applications.

SELECTED PUBLICATION

Unified Parameter-Efficient Adaptation Framework for Quantized LLMs *2025*
First Author, IJNSEI

- Developed a unified parameter-efficient fine-tuning framework integrating LoRA, LoRA+, and QLoRA under a constrained optimization view for 4-bit quantized decoder-only LLMs, enabling billion-parameter model adaptation on commodity GPUs.
- Built figure-ready, reproducible PyTorch benchmarking environments for controlled comparisons across trainable parameters, memory footprint, and task performance; demonstrated comparable adaptation quality with approximately 1% trainable parameters.

RESEARCH EXPERIENCE

The Wharton School *Jun 2026 - Present*
Research Assistant, Prof. Jonah Berger

- Operationalized curiosity and information-gap theory into a reproducible behavioral-data annotation framework, developing a calibrated 7-point labeling schema, explicit decision rules, edge-case taxonomy, and adjudication logic; refined borderline cases to improve consistency and reproducibility of downstream analysis.

NYU School of Global Public Health *Sep 2025 - May 2026*
Biostatistics Research Assistant

- Designed child/family feedback surveys and a small-N ($N < 200$) tracking framework for extended pediatric psychological evaluations, translating clinical questions into structured variables for analysis.
- Structured binary, ordinal, categorical, and qualitative fields for interpretable comparisons of length of stay by diagnosis (ODD/ADHD/depression) and evaluation type (firesetting, PSB, violence risk), explicitly accounting for small-sample limits on inference.

NYU School of Global Public Health *Mar 2025 - May 2026*
Research Assistant, Public Health Backlash Lab

- Collaborated with a cross-institutional NYU/Ohio State/University of Minnesota team to integrate quantitative trends with qualitative evidence on public-health backlash and its policy implications.
- Conducted Dedoose coding and analytic memoing and synthesized mixed-methods findings on communication, policy response, community reaction, and public trust.

TECHNICAL INDUSTRY EXPERIENCE

ByteDance (TikTok) *Jun 2026 - Aug 2026*
AI Product Manager Intern - Search Ads

- Designed and evaluated query-similarity routing and dynamic-threshold strategies across recall, ad load, and auction quality, contributing more than \$145K in incremental revenue while preserving organic-search relevance.
- Prototyped an LLM-powered Q&A advertising experience and defined evaluation logic linking answer-seeking intent, query relevance, generated-content quality, conversion, and overall search quality.

ThunderSoft *Jun 2025 - Jul 2025*
Algorithm Intern

- Built a 100K+ sample annotation/QC pipeline for a Xiaomi mobile-AI project, defining taxonomy and quality-control rules and training three annotators to improve dataset consistency.
- Fine-tuned PyTorch models and ran reproducible latency/accuracy benchmarks, improving accuracy by approximately 8%, reducing inference latency by approximately 5%, and increasing evaluation efficiency by approximately 20%.

ENTREPRENEURSHIP & LEADERSHIP

HaoXue AI Learning Center *May 2023 - Present*
Founder

- Founded and scaled an offline AI-powered learning and enrichment center serving 500+ students; led program design, student growth, and parent engagement.

UNICEF at NYU *Mar 2025 - May 2026*
Founder & President

- Founded NYU's first UNICEF student chapter, led a 20-person team, and delivered programming including a 150+ attendee flagship event.